

JavaScript JSON Data Types

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JSON Value Types

In JSON, values must be one of the following data types:

- a **string**
- a **number**
- an **object** (JSON object)
- an **array**
- a **boolean**
- **null**

JSON values **cannot** be one of the following data types:

- a **function**
- a **date**
- **undefined**

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript JSON Data Types</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>JSON Value Types</h4>
<p id="demo"></p>

<script>
// JSON with different value types
const jsonString = '{"name":"John","age":30,"isStudent":false,"car":null,"hobbies":["reading","m
const obj = JSON.parse(jsonString);

document.getElementById("demo").innerHTML =
  "String: " + obj.name + "<br>" +
  "Number: " + obj.age + "<br>" +
  "Boolean: " + obj.isStudent + "<br>" +
  "Null: " + obj.car + "<br>" +
  "Array: " + obj.hobbies;
</script>

</body>
</html>
```

Example 1

Result [View Example](#)

JSON String Values

In JSON, string values must be written with double quotes:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript JSON Data Types</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>JSON String Values</h4>
<p id="demo"></p>

<script>
// JSON with string values
const jsonString = '{"firstName":"John","lastName":"Doe","city":"New York"}';
const obj = JSON.parse(jsonString);

document.getElementById("demo").innerHTML =
  "First name: " + obj.firstName + "<br>" +
  "Last name: " + obj.lastName + "<br>" +
  "City: " + obj.city;
</script>

</body>
</html>
```

Example 2

Result [View Example](#)

JSON Number Values

In JSON, number values must be an integer or a floating point. They must not be written in quotes:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript JSON Data Types</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>JSON Number Values</h4>
<p id="demo"></p>

<script>
// JSON with number values (integers and floating point)
const jsonString = '{"age":30,"weight":75.5,"height":180}';
const obj = JSON.parse(jsonString);

document.getElementById("demo").innerHTML =
  "Age: " + obj.age + " (integer)<br>" +
  "Weight: " + obj.weight + " (float)<br>" +
  "Height: " + obj.height + " cm";
</script>

</body>
</html>
```

Example 3

Result [View Example](#)

JSON Boolean Values

In JSON, boolean values must be either `true` or `false` (without quotes):

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript JSON Data Types</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>JSON Boolean Values</h4>
<p id="demo"></p>

<script>
// JSON with boolean values
const jsonString = '{"isStudent":false,"hasGraduated":true,"isEmployed":true}';
const obj = JSON.parse(jsonString);

document.getElementById("demo").innerHTML =
  "Is student: " + obj.isStudent + "<br>" +
  "Has graduated: " + obj.hasGraduated + "<br>" +
  "Is employed: " + obj.isEmployed;
</script>

</body>
</html>
```

Example 4

Result [View Example](#)

JSON Null

In JSON, `null` represents an empty or non-existent value:

```
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript JSON Data Types</h2>
<p>From w3schools.com, Experiment by Teeratus_R</p>

<h4>JSON Null</h4>
<p id="demo"></p>

<script>
// JSON with null values
const jsonString = '{"firstName":"John","middleName":null,"lastName":"Doe"}';
const obj = JSON.parse(jsonString);

document.getElementById("demo").innerHTML =
  "First name: " + obj.firstName + "<br>" +
  "Middle name: " + obj.middleName + " (null)<br>" +
  "Last name: " + obj.lastName;
</script>

</body>
</html>
```

Example 5

Result [View Example](#)

Document

Document in project

You can [Download PDF](#) file.

Reference

- [W3Schools JavaScript JSON Data Types](#)